



PARASITIC DISEASES — FULL MIND MAP

Biology Ch. 9.iv · Diseases & Treatment · Hinglish · SSC / BPSC / BSSC / Railway



Topics in this sub-chapter:

1. Malaria — Organ, Vector & Protozoan Basics
2. Malignant Malaria & Plasmodium Species
3. Malaria Discovery, Treatment & Vaccines
4. Filariasis — Nematodes & Mosquito Vectors
5. Disease–Pathogen / Vector Matching
6. Sleeping Sickness — Trypanosoma & Tse-tse Fly
7. Unsafe Water, Sanitation & Communicable Diseases
8. Taeniasis & Neurocysticercosis

1. Malaria — Organ, Vector & Protozoan Basics

- Malaria Plasmodium parasite se linked hai aur source female Anopheles mosquito ko iska carrier batata hai. Malaria mein spleen (pleeha) enlarge ho sakti hai aur source ise most affected organ batata hai; Quinine traditional treatment drug ke roop mein mention hai. [Q1, Q2, Q3]
- Source ke according Anopheles mosquitoes malaria ke alawa kuch regions mein encephalitis aur filariasis ke vectors bhi ho sakte hain; filariasis ke other important vectors mein Culex aur Aedes bhi mention hain. [Q4]
- Malaria aur Dengue dono mein fever aur mosquito-bite transmission common context hai, lekin mosquito species same nahi: Malaria → female Anopheles; Dengue → female Aedes aegypti. [Q7]
- Malaria protozoan disease hai: causative parasite = Plasmodium. Isliye “insect parasite se caused” statement source question mein false hai; source mosquitoes se spread, swampy-region association aur chloroquine treatment ko correct statements batata hai. [Q8, Q9, Q10]

2. Malignant Malaria & Plasmodium Species

- Malignant/severe malaria ka causative species source mein Plasmodium falciparum diya hai; ye severe human infectious disease aur higher death-risk se linked hai. [Q5, Q6]
- Source ke mutabik Plasmodium vivax recurring malaria ka frequent/widely distributed cause hai aur India ke about 65% malaria cases ke liye responsible bataya gaya hai; Asia aur South America ke context mein bhi 65% figure mention hai. [Q11]

3. Malaria Discovery, Treatment & Vaccines

- Source Q12 ka answer Sir Ronald Ross hai; explanation mein British Indian Army doctor ke roop mein female Anopheles mosquito mein Plasmodium oocysts observe karna aur malaria work ke liye 1902 Nobel Prize milna mention hai. [Q12]
- R21/Matrix-M malaria vaccine Jenner Institute, Oxford University aur Serum Institute of India collaboration se developed batayi gayi hai, Novavax saponin-based adjuvant technology ke saath. Source ke according October 2023 mein WHO recommendation mili aur ye children ke use ke liye second malaria vaccine hai; RTS,S/AS01 (Mosquirix) first WHO-approved malaria vaccine diya hai. [Q13]



EXAM TRAP — Source-Specific Malaria Key

- ▶ Q12 ka wording “malaria is caused by Plasmodium parasite” ki discovery Sir Ronald Ross se map karta hai; mind map source ke marked answer aur explanation ko exactly preserve karta hai. [Q12]

4. Filariasis — Nematodes & Mosquito Vectors

- Filariasis thread-like nematodes/filariae se caused parasitic infection hai. Source female Culex mosquito se person-to-person spread, edema ko common symptom aur severe swelling/thickened skin ko elephantiasis se link karta hai. [Q14]

- Vector–Disease–Pathogen set mein Aedes–Chikungunya–Bacteria incorrect hai, kyunki source Chikungunya ko Aedes-transmitted viral disease batata hai. Culex–Lymphatic Filariasis–Parasite, Anopheles–Malaria–Parasite aur Aedes–Dengue–Virus correct sets hain. [Q15]

5. Disease–Pathogen / Vector Matching

- Correct match: Malaria → Protozoan; Poliomyelitis → Virus; Tuberculosis → Bacteria; Ringworm → Fungi. [Q16]
- Source mapping: Plague → Bacteria; AIDS → Virus; Baldness → Fungus; Malaria → Protozoa. [Q17]
- Another source match: Peptic ulcers → Bacteria; Dengue → Virus; Elephant Foot → Worm; Malaria → Protozoa. [Q18]

EXAM TRAP — Source-Specific Matching

- ▶ Q17 mein source “Baldness → Fungus” mapping deta hai; isi source-key ko yahan preserve kiya gaya hai. [Q17]

6. Sleeping Sickness — Trypanosoma & Tse-tse Fly

- Sleeping sickness / human African trypanosomiasis Trypanosoma brucei rhodesiense aur T. brucei gambiense jaise unicellular parasites se caused hoti hai aur infected tse-tse fly (Glossina) ke bite se spread hoti hai. [Q19]
- Sleeping sickness mosquito-borne disease nahi hai. Source comparison mein Dengue, Filariasis aur Malaria mosquito-borne diye gaye hain, jabki sleeping sickness tse-tse fly se transmit hoti hai. [Q20]

7. Unsafe Water, Sanitation & Communicable Diseases

- Source question unsafe drinking water aur lack of sanitation ke context mein developing countries ki three main communicable diseases ko Malaria, Acute diarrhoea aur Schistosomiasis ke roop mein list karta hai. [Q21]

8. Taeniasis & Neurocysticercosis

- Taeniasis intestinal tapeworm infection hai. Source three human species deta hai: Taenia solium (pork tapeworm), Taenia saginata (beef tapeworm) aur Taenia asiatica (Asian tapeworm); major health problems especially T. solium se linked hain. [Q22]
- T. solium eggs/larval cysts (cysticerci) various organs mein develop ho sakte hain; central nervous system, especially brain/spinal cord, involve hone par neurocysticercosis (NCC) aur neurological symptoms including epileptic seizures ho sakte hain. [Q22]
- Leander Paes 2003 case mein source brain problem ka answer “a cyst” deta hai; explanation common Indian brain lesion ko cysticercus granuloma—tapeworm ke larval form—se link karti hai. [Q23]